

1 DATA

PhysioNet EEGMMIDB
104 subjects, 45 trials

BCI IV-2a
9 subjects, 288 trials
two sessions

Cho2017
52 subjects, 200 trials

the same 8 sensorimotor channels, 8 to 30 Hz, one-second window
spatial covariance C (8×8 , symmetric positive definite)

2 FRAME

EEG nuisances (electrode gain, head geometry, source mixing) act on C by congruence $C \rightarrow ACA^T$

sensor frame
 $\rho = C / \text{tr} C$
invariant under OCO^T only

reference frame
 $\tilde{\rho} = WCW / \text{tr}(WCW)$, $W = M^{-1/2}$
exactly invariant under ACA^T

every suite is run in both frames, changing nothing else

3 PIPELINES

classical baselines

CSP + LDA
TS + LR
MDM
FBCSP

metric-matched twin

Riemannian
kernel SVM
(log-Euclidean
as a second)

quantum kernels

HS overlap
Fidelity
HS-RBF
Bures-RBF
QRE-RBF

circuit kernels + ablation

IQP entangled
IQP entanglers
deleted
PCA-matched

**same covariances, same SVM, same tuning budget, same frame:
only the metric differs**

4 EVALUATION

nested cross-validation: 5 folds $\times$ 3 repeats outer, 4-fold inner grid search,
the identical search budget for every pipeline, quantum and classical

paired per-subject Wilcoxon signed-rank, Holm corrected within each pre-specified family,
and two one-sided tests for equivalence against the twin